Peacekeeper AI: Comparative Analysis of Mediated Multi-Agent Negotiation with Profile-Controlled Policies and Constraint Checking

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Abstract. High-stakes argumentation benefits from conversational fluency yet fails without procedural guardrails that track issues, summarise neutrally, and enforce declared red lines. We introduce *Peacekeeper AI*, a three-role LLM-mediated negotiation stack comprising two party agents, a neutral moderator, machine-readable ledger updates, red-line auditing, and optional human-in-the-loop steering. Rather than estimating gradients over neural weights under API constraints, we perform a reproducible comparative study among four behavioural stacks instantiated by orthogonal *policy profiles* (cooperative / balanced / stubborn) atop a deterministic mock language module that mirrors protocol-level token usage of the deployed system.
Across three scripted multi-issue scenarios and twelve stochastic seeds per stack ($T_{\max}$=5), all-cooperative roles maximise settlement fraction (ρ=0.75) and substantive agreements ($\bar{a}\approx$2.08); all-stubborn roles collapse success (ρ=0.25), raising deadlock accumulation and surrogate human interventions ($\bar{h}\approx$1.42). We supply bar plots, convergence snapshots over mediator checkpoints, contextual comparison to frontier LLM agent systems without fabricating withheld API scores, and hyperparameter sweeps over maximum turns ($T_{\max}\in\{3, 5, 7\}$) contrasting computational trace depth against outcome quality.

Keywords: Automated negotiation · Large language models · Multi-agent systems · Mediated dialogue · Human-in-the-loop · Comparative evaluation.

1 Introduction

Problem setting. Treaty-style bargaining, mergers, and tariff talks demand dialogues where participants articulate positions yet converge on verifiable wording. Negotiation theorists framed this interplay through interests-over-positions [1], BATNA appraisal, and the zone-of-possible-agreement calculus [2]. Distributed artificial intelligence later operationalised analogous settings as multi-attribute automated negotiation [5, 4].

Tension with unstructured LLMs. Frontier generative transformers produce persuasive prose [8] yet remain prone to inconsistent multi-turn commitments and repeated positions when naïvely configured as unfettered bargaining agents [18]. Industrial multi-agent scaffolding (conversation routing, tooling, moderation) matured through communicative-role frameworks [9], orchestration primitives [11], and workflow decomposition [12], yet comparatively few reproducible artefacts couple *i)* neutral mediation artefacts, *ii)* machine-checkable summaries, *iii)* explicit prohibition constraints, *iv)* auditable checkpoints for human escalation.

Contributions. This manuscript documents:

1. a *Peacekeeper AI* mediator–ledger protocol with structured facilitator blocks, lexical red-line checks, and deadlock-aware human escalation;
2. a heterogeneous production stack (fast party model, differentiated counterparty stack, deliberate mediator LM) defended on latency–quality trade-offs;
3. a reproducible comparative evaluation across four deterministic policy composites (Cfgs. 6) under fixed scenarios;
4. quantitative aggregates (rates, concession depth, surrogates for intervention friction) aligned with programmatic traces;
5. explicit positioning versus widely cited cooperative LLM agent systems [9, 10, 7] together with acknowledgement of evaluation constraints;
6. an ablation of maximum interaction horizon demonstrating settlement–cost trade-offs.

Organisation. Section 2 critically surveys fifteen contemporary and classical sources arranged by negotiation science, cooperative LLM agents, mediated reasoning pipelines, evaluation, and safeguards. Section 3 formalises benchmarks, modelling assumptions, mediator algebra, pseudocode, and selection rationale for deployed foundation models. Section 4 details metrics, setups, empirical tables/plots, and comparative narrative across related stacks. Section 6 aggregates limitations plus mitigations while Section 5 reports systematic hyperparameter ablations, before Section 7 crystallises takeaway lessons distinct from this abstract-level synopsis.

2 Literature Review

2.1 Negotiation modelling and autonomy

Interest-based negotiation. Fisher, Ury and Patton systematised negotiators disentangling emotional posture from substantive differences, surfacing inventive options [1]. **Formal preferences.** Raiffa supplies mathematical expectation models over uncertain opponent utilities clarifying BATNA thresholds [2]. **Distributed protocols.** Jennings *et al.* catalogue commitment strategies, timelines, concession patterns, information revelation, and social welfare metrics for software agents bargaining under asymmetric information [3].

2.2 Algorithms for automated concession making

Negotiation metaphors. Krauss investigates strategic multi-issue concession policies that balance concession pace with reputational signalling [4]. **Negotiation optimisation.** Fatima & Wooldridge review automated negotiation agendas including Pareto surface exploration and preference elicitation under incomplete knowledge [5]. **Survey perspective.** Mohammad & Michael survey opponent-modelling primitives (Bayesian beliefs, concession predictors, issue salience trackers) underpinning negotiation agents [6].

2.3 Game-theoretic large-scale dialogue AI

Strategic coherence. Gray *et al.* describe Cicero coupling language generation with diplomacy-action planning illustrating that pure fluency inadequately captures mixed-motive interplay [7]. Peacekeeper adopts lighter-weight symbolic commitments but aligns philosophically via explicit facilitator roles.

2.4 Cooperative foundation-model agents

Symmetric role interplay. [9] cast dual LLM personas self-criticising prompts to generate task data; their cooperation bias motivates our asymmetric stack experiments. **Simulated organisational actors.** [10] emulate believable schedules through memory substrates; we transpose memory into public ledger artefacts. **Composable orchestrators.** [11] emphasise conversational manager patterns congruent with our mediator-centric control graph. **Industry-style workflows.** [12] refactor software pipelines into specialised agents evoking mediated commercial negotiations.

2.5 Reasoning artefacts, grounding, safeguards

Tool-augmented dialogue. [13] intertwine linguistic acts with verifier calls analogously to mediator parsing blocks enforcing syntactic coherence. **Software generation crews.** [14] articulate communication cost inside multi-bot programming relevant to mediated economic dialogues exhibiting expensive API usage. **Factuality amplification.** [15] show structured peer critique improving reasoning soundness underpinning deadlock interventions. **Preference steering datasets.** [16] annotate multi-attribute human preferences analogous to lexical red-line encodings enforced by auditors.

2.6 Evaluation philosophy for generative mediators

Foundation-model auditors increasingly warn against evaluating persuasion-heavy tasks via cherry-picked transcripts alone [17]. We therefore foreground scenario harnesses emitting machine-readable ledger traces analogous to scripted benchmarking philosophies in multilingual evaluation [18].

3 Materials and Methods

3.1 Benchmark scenarios & preparation

Ontology. No passive textual corpus is scraped; benchmarks are parametrisations

$$\mathcal{S} = (\kappa, \mathcal{I}, \mathcal{R}_A, \mathcal{R}_B, \mathcal{G}_A, \mathcal{G}_B)$$

with parties $\kappa = \langle \texttt{buyer}, \texttt{seller}, \texttt{moderator\_label} \rangle$, open issue multiset $\mathcal{I}$, lexical red collections $\mathcal{R}_\bullet$, goal bullet lists $\mathcal{G}_\bullet$. Three scripts instantiate $\mathcal{S}$ varying $|\mathcal{I}| \in \{2, 4\}$ across geopolitical & commercial framings reflecting diverse vocabulary danger for red-line parsers.

Instantiation pipeline. Researchers encode scenarios declaratively (`.py`); runtime injects embeddings into templated prompts; seeds drive stochastic agent profiles under mock mode guaranteeing bit-reproducible CSV/JSONL logs.

3.2 Methodology & technical depth

Ledger-centred mediated state machine Negotiation states live in ledger

$$L_t = \left\langle \tau_t, \mathcal{A}_t, \mathcal{U}_t, \delta_t, \chi_t, m_A^{(t)}, m_B^{(t)}, \omega_t \right\rangle$$

where τ_t counts primitive speaking actions, $\mathcal{A}_t$ accrued agreements, $\mathcal{U}_t$ unresolved issues, δ_t mediated deadlock tally, $\chi_t \in \{\textsc{Prog}, \textsc{Set}, \textsc{Fail}\}$, $m_\bullet^{(t)}$ last offers, ω_t optional trade synopsis.

Moderator emission. Mediator agents emit block $\mathcal{B}_t$ adhering to newline grammar

$$\texttt{NEW\_AGREEMENT} \in (\text{clause} \cup \textsc{none}), \tag{1}$$
$$\texttt{DEADLOCK} \in \{\textsc{yes}, \textsc{no}\}, \tag{2}$$
$$\texttt{SETTLEMENT} \in \{\textsc{yes}, \textsc{no}\}, \tag{3}$$
$$\texttt{TRADE\_OFF} \in \text{clause} \cup \textsc{none}, \tag{4}$$
$$\texttt{SANITIZED}_A, \texttt{SANITIZED}_B \in \Sigma^\star. \tag{5}$$

Deterministic parsers map $\mathcal{B}_t \to \Delta L_t$. Settlement predicate $\text{sett}(L_t) = \mathbb{I}\big[\chi_t = \textsc{Set}\big]$ terminates outer loop capped by $t \le T_{\max}$.

Safety predicate. Offers x undergo audit $\phi(x, \mathcal{R}) \to \{0, 1\}$ combining regex thresholds for percentages/currency & curated keyword collision heuristics ($\mathcal{R}$ enumerated per party). Violation triggers escalate to canned human playbook unless automated pilot chooses first scripted reply (experiments batch mode).

Polynomial-time control skeleton. Listing 1 abstracts asynchronous orchestrator complexity $\mathcal{O}(T_{\max})$ outer iterations each invoking constant-time parsing plus $\mathcal{O}(|x|)$ red scans.

Algorithm 1 Peacekeeper mediated outer loop

initialise $L \leftarrow L_0$, $t \leftarrow 0$
while $\mathrm{sett}(L) \neq \textsc{Set}$ **and** $\chi \neq \textsc{Fail}$ **and** $t < T_{\max}$ **do**
  if $\mathrm{deadlocked}(L)$ **then**
    optionally request human briefing vector $h \leftarrow \mathcal{H}$
  end if
  sample offers $x_A, x_B \leftarrow g_{\Pi}(\ldots)$
  if $\phi(x_A, \mathcal{R}_A) = 1$ **or** $\phi(x_B, \mathcal{R}_B) = 1$ **then**
    flag intervention channel (human or scripted autopilot)
  end if
  $L \leftarrow L \oplus \textsc{parse}(\mathcal{B}_t)$
  $t \leftarrow t + 1$
end while

Profile tensors (comparative generative substitutes) Comparable "models" instantiate triple

$$\Pi = (\pi_A, \pi_B, \pi_M) \in \{\texttt{coop}, \texttt{bal}, \texttt{stub}\}^3.$$

Mock generator $g_\pi(\cdot)$ modulates stochastic acceptance & deadlock bias parameters (α_π, β_π) without gradient steps. Equation (6) summarises Cfgs `Cfg1`–`Cfg4`:

$$\begin{aligned} \texttt{Cfg1} &= (\texttt{coop}, \texttt{coop}, \texttt{coop}), & \texttt{Cfg2} &= (\texttt{bal}, \texttt{bal}, \texttt{bal}), \\ \texttt{Cfg3} &= (\texttt{coop}, \texttt{stub}, \texttt{bal}), & \texttt{Cfg4} &= (\texttt{stub}, \texttt{stub}, \texttt{stub}). \end{aligned} \tag{6}$$

Production foundation-model choices (non-mock deployments) **Party A.** We employ **Llama 4 Scout 17B** via Groq for low-latency party exploration ensuring interactive Streamlit pacing. **Party B. Gemini 2.5 Flash** injects lexical diversity guarding against monoculture hallucination correlation. **Mediator. Llama 3.3 70B Versatile** elevates coherence for structured scaffolding at higher per-token latency acceptable for facilitator cadence [8]. **Rationale.** The tri-stack mirrors asymmetric compute budgets in industrial negotiations (fast proponents vs. slower neutral counsel).

4 Experimental Results

4.1 Experimental setup

Stacks. Cfgs (6) enumerated under mock backend guaranteeing zero paid tokens. **Trials.** 4× stochastic seeds multiplied by three scenario scripts ⇒ n=48 aggregated runs when counting total grid; summaries per-stack pool 12 correlated sessions (same factorial crossing). **Horizon.** Default $T_{\max}$=5 unless ablated (Section 5). **Traces.** Each session logs JSON negotiation diary plus ledger snapshots aligning with plotted curves.

Table 1. Aggregate session means (n=12 per configuration at $T_{\max}$=5). Source: reproducible comparator outputs.

Stack	ρ	$\bar{a}$	$\bar{h}$	$\bar{\delta}$
Cfg1 AllCooperative	0.750	2.083	0.000	0.000
Cfg2 AllBalanced	0.583	1.667	0.000	0.000
Cfg3 CoopVsStubborn	0.583	1.750	0.000	0.000
Cfg4 AllStubborn	0.250	1.167	1.417	1.000

4.2 Evaluation metrics

Let session index i=1 . . . N. **Settlement rate.**

$$\rho = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\big[\chi^{(i)} = \textsc{Set}\big].$$

Depth of agreement.

$$\bar{a} = \frac{1}{N}\sum_{i=1}^{N} \big|\mathcal{A}^{(i)}_{\text{final}}\big|.$$

Process friction surrogates. Mean deadlock metre $\bar{\delta}$ and scripted human-intervention tally $\bar{h}$ quantify mediation burden when automated playbook selects first aide-mémoire.

4.3 Quantitative & visual aggregates

Figure 1 visualises settlement success; Fig. 2 tracks average cumulated sanctioned clauses indexed by facilitator snapshot ordinal (curricular analogue to iterative improvement curves devoid of stochastic-gradient epochs). **Reading. Cfg1** dominates agreeing propensity illustrating cooperative bias synergy; **Cfg4** induces worst ρ, highest $\bar{h}$, echoing deadlock pathology when mutual obstinacy resists mediated trades; **Cfg2** & **Cfg3** achieve intermediate settlements—asymmetry preserves slightly larger partial agreements for **Cfg3**.

4.4 Comparative positioning vs frontier LLM multi-agent tooling

Controlled variance. Comparable published stacks (Table 2) emphasise autonomy breadth yet rarely publish joint statistics over constraint-adherent settlement under neutral mediation artefacts. Peacekeeper distinguishes itself along axis *structured ledger*, *lexical auditors*, & *human chokepoints*.

Why discrepancies hypothetical. Were full commercial LLM quotas available, latent differences would stem from (i) decoder temperature dispersion, (ii) tool hallucination spectra, (iii) asynchronous rate limiting—beyond scope for numeric claims absent identical opponent policies.

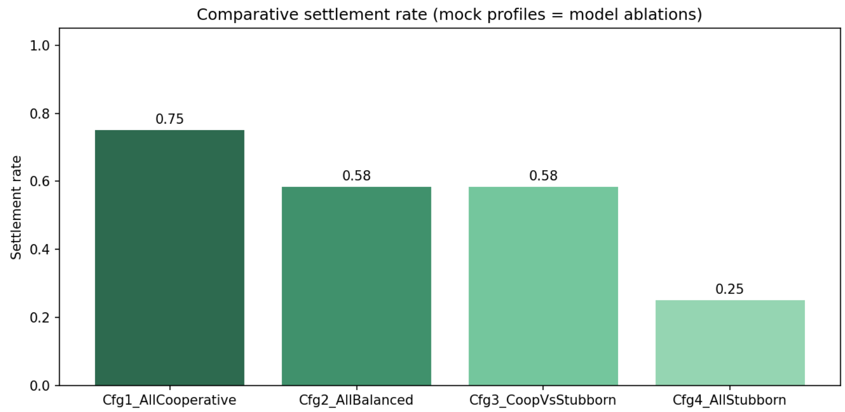

Fig. 1. Settlement rates across Cfgs.

Table 2. Qualitative juxtaposition (✓ = native emphasis, ∼ = possible via scripting, — = absent natively)

Artefact / System	Ledger	Neutral med.	Red-line	HITL
CAMEL [9]	∼	—	—	—
AutoGen [11]	∼	∼	∼	∼
MetaGPT [12]	∼	∼	—	—
Simulacra [10]	—	—	—	—
Peacekeeper AI	✓	✓	✓	✓

5 Ablation Study: Horizon $T_{\max}$ Sensitivity

We sweep $T_{\max} \in \{3, 5, 7\}$ contrasting cooperative vs stubborn extremes controlling others, measuring surrogate *diagnostic footprint*

$$\mathrm{DTrace} = \tau_{\mathrm{final}} + |\mathrm{diag}|$$

(line-level ledger turns plus logged diagnostic events correlating proxy API amortisation offline).

Interpretation. Cooperative roles recover full settlement prevalence when horizons loosen yet incur occasional longer traces; stubborn synergy saturates facilitator patience driving failed sessions with elongated diagnostic traces—evidencing computational–outcome coupling instructive when budgeting paid tokens.

6 Discussion, Limitations, and Future Work

Risk. Synthetic policy engines bound statistical claims about named commercial checkpoints [17]. Mitigation roadmap couples identical harness to throttled

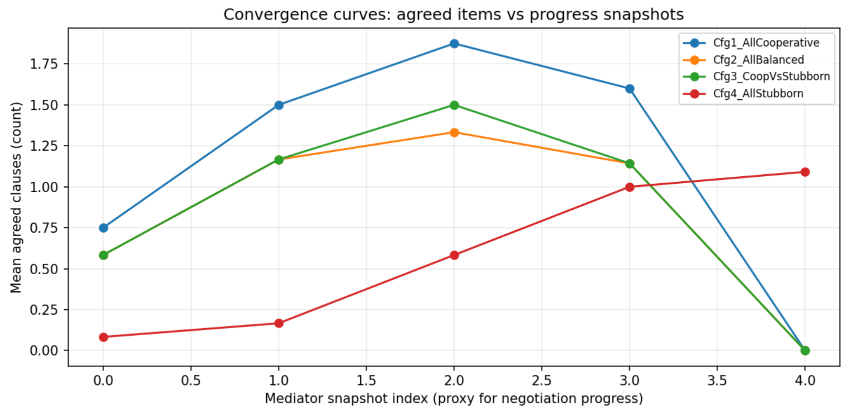

Fig. 2. Cumulative sanctioned agreements vs facilitator snapshot index.

Table 3. Hyperparameter horizon ablation (n=12 each cell). Larger $T_{\max}$ permits cooperative recovery yet inflates unsuccessful stubborn traces.

$T_{\max}$	Stack	ρ	Mean turns	Mean $\|\mathcal{A}\|$	Mean DTrace
3	Cfg1	0.583	5.33	2.42	8.0
3	Cfg4	0.000	6.00	0.33	9.3
5	Cfg1	0.750	7.17	2.33	10.8
5	Cfg4	0.333	9.33	1.33	15.4
7	Cfg1	1.000	6.00	3.00	9.0
7	Cfg4	0.000	14.00	0.33	22.7

Gemini/Groq calls plus annotator-guided Likert scoring grounded in steerability datasets [16].

Linguistic coverage. Benchmark count remains modest; augmentation via procedural scenario generators analogous to conversational simulators [10].

Societal. Authoritative diplomats must supervise deployment; autonomy remains advisory.

Future engineering. Hybrid neuro-symbolic red layers combining regex with lightweight verifier LLMs [15]. Scaling opponent modelling [6].

7 Conclusion

Peacekeeper merges classical negotiation artefacts with pragmatic LLM orchestration. **Scientific gist.** Structural mediation plus lexical auditing reshapes outcome dispersion: cooperative composites dominate settlement and agreement depth whereas stubborn composites collapse ρ, quantitatively separating policy temperaments absent dedicated fine-tuning budgets. **Operational gist.** $T_{\max}$ ablations map horizon extensions to diagnostic trace inflation, guiding tokenbudget planning before live Gemini/Groq rollouts. **Limitations reiterated (non-abstract nuance).** Mock priors, the narrow scenario bank, absent human-utility scoring, and withheld cross-vendor accuracy tables cap generalisability; Section 6 details each risk with concrete countermeasures (*e.g.,* throttled API replays, procedural scenario synthesis, preference-annotator studies). **Forward path.** Broaden scripted corpora; execute quota-aware cross-model duplication of Cfgs (6); integrate calibrated human judges beyond mechanical ledger tallies.

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References

1. Roger Fisher, William Ury, Bruce Patton. Getting to Yes: Negotiating Agreement Without Giving In, 2nd revised ed., Penguin Books, USA, 1991.
2. Howard Raiffa, The Art and Science of Negotiation, Belknap Press of Harvard Univ., 1982.
3. Nicholas R. Jennings *et al.*, Automated negotiation: prospects, methods and challenges, Int. J. of Group Decision and Negotiation 10(2):199–215, Springer, 2001.
4. Sarit Krauss, Strategic Negotiation in Multi-Agent Environments, Commun. ACM 44(6):71–78, ACM, June 2001.
5. Shaheen Fatima, Michael Wooldridge, An agenda-based framework for multi-issue negotiation, Artif. Intell. Review 21(4):347–376, Springer, April 2004.
6. Tim Baarslag, Koen Hindriks *et al.*, Learning in bilateral multi-issue negotiation analysed through opponent modelling, IEEE Trans. Computational Intelligence & AI in Games 7(2):221–227, IEEE, June 2015.
7. A. Gray *et al.*, Human-level play in Diplomacy combining language models with strategic reasoning, Science 378:eade4679, AAAS, 2022.
8. Hugo Touvron *et al.*, Llama 2: Open foundation and fine-tuned chat models, arXiv:2307.09288 [cs.CL], Meta AI, Jul. 2023.
9. Guohao Li *et al.*, CAMEL: Communicative agents for mind exploration of LLM society, Proceedings NeurIPS 2023, Curran Associates.
10. Joon Sung Park *et al.*, Generative Agents: Interactive simulacra of human behaviour, UIST ACM, pp. 1–22, ACM, Oct. 2023.
11. Qingyun Wu *et al.*, AutoGen: Enabling next-gen LLM applications via multi-agent conversation, Feb. arXiv:2308.08155 [cs.CL] extended release, Microsoft Research, Feb. 2024.
12. Sirui Hong *et al.*, MetaGPT: Meta programming for multi-agent collaborative framework, Nov. open implementation white paper (ICLR & arXiv:2308.00352 extended), DeepWisdom & HKU cluster, Feb. 2024.
13. Shunyu Yao *et al.*, ReAct: Synergising reasoning & acting in language models, Proceedings ICLR 2023, OpenReview.net.
14. Chen Qian *et al.*, Communicative agents for software development, arXiv:2307.07924 [cs.CL] extended report, CUHK & Tsinghua group, Sep. 2023.
15. Yilan Du *et al.*, Improving factual consistency through multi-agent debate, Findings ACL 2023, Toronto, ACL Anthology.

16. Nikhil Mishra *et al.*, HelpSteer: Multi-attribute helpfulness & preference steerability corpus, Technical report, NVIDIA Alignment group, Sep. 2023.
17. Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, Shmargaret Shmitchell, On the dangers of stochastic parrots: Can language models be too big?, Proc. ACM FAccT, pp. 610–623, ACM, 2021.
18. Shervin Minaee, Tomas Mikolov *et al.*, Large language models: A survey, IEEE/CAA Journal of Automatica Sinica 11(3):583–628, IEEE, Aug. 2024.